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BRIEF NOTES

Heaviside projection based topology optimization by a PDE-filtered scalar function

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Abstract This paper deals with topology optimization based on the Heaviside projection method using a scalar function as design variables. The scalar function is then regularized by a PDE based filter. Several image-processing based filtering techniques have so far been proposed for regularization or restricting the minimum length scale. They are conventionally applied to the design sensitivities rather than the design variables themselves. However, it causes discrepancies between the filtered sensitivities and the actual sensitivities that may confuse the optimization process and disturb the convergence. In this paper, we propose a Heaviside projection based topology optimization method with a scalar function that is filtered by a Helmholtz type partial differential equation. Therefore, the optimality can be strictly discussed in terms of the KKT condition. In order to demonstrate the effectiveness of the proposed method, a minimum compliance problem is solved.

Keywords Topology optimization · Heaviside projection method · Filter

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1 Introduction

In continuum-based topology optimization, image-processing techniques are commonly used for filtering oscillatory designs or controlling a minimum length scale (Bendsøe and Sigmund 2003). They are categorized into two major groups, namely density filters (Bourdin 2001; Bruns and Tortorelli 2001) and sensitivity filters (Sigmund 2007). The sensitivity filters are applied to the design sensitivities rather than the design variables themselves. They have been popular because one can overlook the strict bounds [0, 1] for the design variables when manipulating the design sensitivities. Among those filters the so-called convolution filter is the most widely used for removing the checker board patterns (Bendsøe and Sigmund 2003). It is simple to implement and easy to understand its geometric effects. One drawback is, however, that there remain discrepancies between the filtered sensitivities and the actual sensitivities. When the filtered sensitivities are fed to a modern optimization solver such as SNOPT (Gill et al. 2007), the false sensitivity values may confuse the optimization process because the optimization algorithm is based on the original function values to fulfill the first order optimality conditions. It is even unclear what optimization problems are to be solved after filtering the design sensitivities. On the other hand, a PDE-based filter has been recently proposed as an alternative sensitivity filter, in which a Helmholtz type partial differential equation is solved instead of the conventional image-processing (Lazarov and Sigmund 2009). By appropriately designing the partial differential equation, a similar convolution filter can be realized in the sense of the associate Green function. From the implementation point of view this filter is advantageous because it can utilize the existing computational framework of FEM. In this paper, we propose to apply the PDE-based filtering technique directly

to the design variables instead of the design sensitivities. To this end we first introduce a scalar function as the design variables. Then, we regularize the scalar function by solving a Helmholtz type partial differential equation with the scalar function values as inputs. Finally, we project the filtered scalar function to the material density by a regularized Heaviside function so that the sensitivity analysis is possible (Guest et al. 2004).

2 Topology optimization

In this section we first present the design representation by using a scalar function and a Heaviside function, then introduce a filter based on a Helmholtz type partial differential equation, and finally present a relevant problem formulation.

2.1 Design representation

Let us assume that a fixed design domain Ω is given, and that the domain consists of a subdomain Ω_s that stands for solid material and the complementary domain for void. Figure 1 shows one instance of a structure. The shape of the target structure can be represented by a characteristic function χ:

$$\chi(\mathbf{x}) = \begin{cases} 0 & \text{for} \quad \forall \mathbf{x} \in \Omega_s \backslash \left(\partial\Omega_s \backslash \partial\Omega\right); \\ c & \text{for} \quad \forall \mathbf{x} \in \partial\Omega_s \backslash \partial\Omega; \\ 1 & \text{for} \quad \forall \mathbf{x} \in \Omega \backslash \Omega_s, \end{cases} \tag{1}$$

where $\mathbf{x}$ stands for a position in Ω, and $\partial\Omega_s$ is the boundary of Ω_s.

The above characteristic function χ can be expressed by the following Heaviside function together with a scalar function ϕ (see also Fig. 2):

$$\chi(\mathbf{x}) = H_c(\phi(\mathbf{x})) = \begin{cases} 0 & \left(\phi(\mathbf{x}) < \phi_0\right); \\ c & \left(\phi(\mathbf{x}) = \phi_0\right); \\ 1 & \left(\phi(\mathbf{x}) > \phi_0\right), \end{cases} \tag{2}$$

where ϕ_0 is the threshold with an intermediate constant value ($0 \leq c \leq 1$); the scalar function ϕ is bounded within $[\underline{\phi}, \overline{\phi}]$. With this expression, one can design arbitrary shapes and topologies of the solid material domain Ω_s in the fixed design domain Ω.

In projection based topology optimization, the scalar function has been conventionally used in the bounds $[\underline{\phi}, \overline{\phi}] = [0, 1]$ as design variables called density. But in fact, they have no physical meaning but only intermediate mathematical variables that can essentially be scaled into an arbitrary interval (Sigmund 2007). In this paper, we will use the interval $[\underline{\phi}, \overline{\phi}] = [-1, 1]$ and the zero threshold as $\phi_0 = 0$ instead.

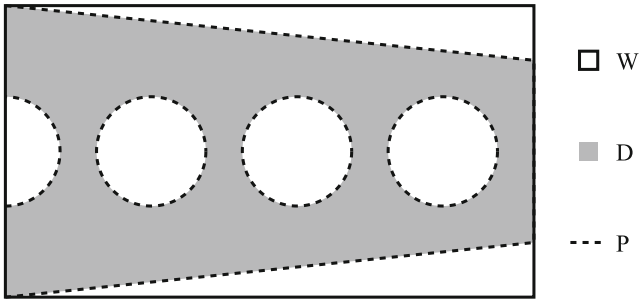


Fig. 1 Fixed design domain and design representation

2.2 Helmholtz PDE based filter

Since the scalar function ϕ ($\in L^\infty(\Omega)$) can take any pointwise value, the raw scalar function may produce severely oscillating designs like a checkerboard or a corrugated shape. In order to regularize the scalar function we introduce the following partial differential equation for $\varphi (\in H^1(\Omega))$:

$$-R^2\nabla^2\varphi + \varphi = \phi. \tag{3}$$

Without the first term this equation simply substitutes ϕ into φ, while the equation becomes Poisson's equation with $R \to \infty$. By appropriately setting up R with an intermediate value, this equation functions like a low pass filter that acts on a raw scalar function ϕ to produce the smoothed scalar function φ. The meaning becomes even clearer when we investigate the shape of the associate Green function (Lazarov and Sigmund 2009). Figure 3 shows the Green function in the case where the domain is a unit disc with all Neumann conditions and $R = 0.1$. φ is calculated by the convolution integral of the Green function multiplied by the input ϕ. In other words, we seek the optimum design within the space of smooth φ restricted by the equation.

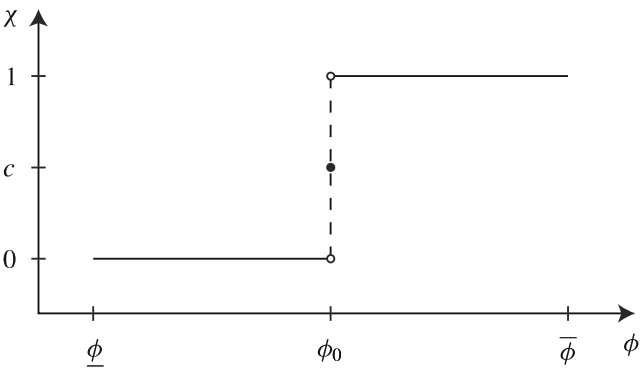


Fig. 2 Heaviside function by (19)

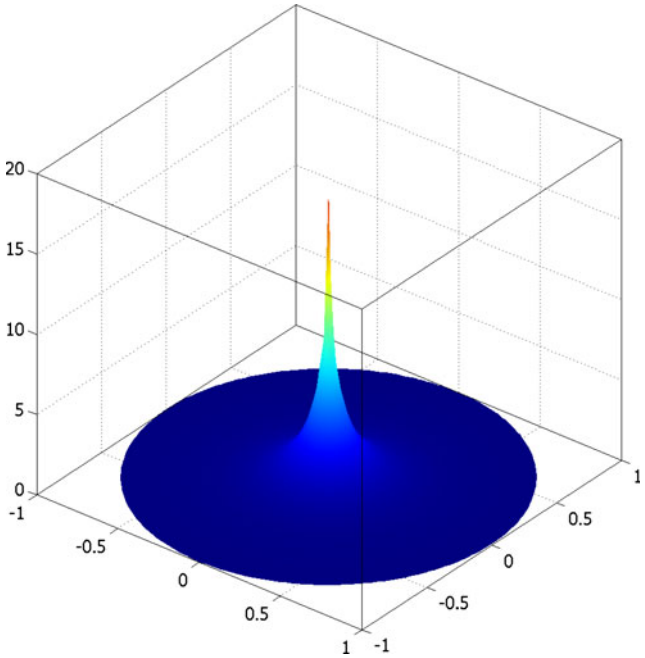


Fig. 3 The shape of the Green function of (3) under the complete Neumann condition

2.3 Material interpolation by a regularized Heaviside function

In order to apply a gradient based optimization technique, the characteristic function χ is relaxed as a material density function that is defined by the following regularized Heaviside function of the filtered scalar function φ:

$$\rho(\varphi) = \begin{cases} \varepsilon & (\varphi < -h) \\ (1-\varepsilon)H(\varphi) + \varepsilon & (-h \le \varphi \le h) \\ 1 & (h < \varphi) \end{cases}, \tag{4}$$

where h is a positive parameter for the bandwidth between the complete material domain (where $h < \varphi$) and the complete void domain (where $\varphi < -h$), therefore, the bandwidth is equal to $2h$. By tightening the bandwidth we can recover the sharpness of the boundary that might be blurred by (3) with large R (Guest et al. 2004). ε is a very small positive lower bound set for avoiding singularity. Here $H(\varphi)$ is defined as

$$H(\varphi) = \frac{1}{2} + \frac{15}{16}\left(\frac{\varphi}{h}\right) - \frac{5}{8}\left(\frac{\varphi}{h}\right)^3 + \frac{3}{16}\left(\frac{\varphi}{h}\right)^5. \tag{5}$$

The derivative of (4) corresponds to the following relaxed Dirac delta function:

$$\frac{\partial \rho(\varphi)}{\partial \varphi} = \begin{cases} 0 & (\varphi < -h); \\ (1-\varepsilon)\delta(\varphi) & (-h \le \varphi \le h); \\ 0 & (h < \varphi), \end{cases} \tag{6}$$

where

$$\delta(\varphi) = \frac{15}{16h}\left\{1 - \left(\frac{\varphi}{h}\right)^2\right\}^2. \tag{7}$$

Since the derivative takes on a non-zero value only in the domain where $\mathbf{x} \in \{\mathbf{x} : -h \le \varphi \le h\}$, $\rho(\varphi)$ can only vary within the bandwidth of $2h$. This means that when the optimization is started at an intermediate value within the bandwidth, the topology radically changes in the early stage, but after the scalar function separating into the bounds, the shape can only change in the final stage.

2.4 Problem formulation

With the above mentioned representation, a minimum mean compliance problem can be formulated as

$$\begin{aligned} \underset{\underline{\phi} \le \phi \le \overline{\phi}}{\text{minimize}} \quad & f := \int_{\Gamma_t} \mathbf{t} \cdot \mathbf{u} \, \mathrm{d}\Gamma \\ \text{subject to} \quad & g := \int_{\Omega} \rho(\varphi) \, \mathrm{d}\Omega - \overline{V} \le 0, \end{aligned} \tag{8}$$

where the design variables are the original scalar function ϕ. The scalar function bounds $[\underline{\phi}, \overline{\phi}]$ make the feasible set of the topology design problem (8) bounded, a property which is essential for a reasonable convergence.

Since the optimization problem (8) is written in the nested form, the displacement vector $\mathbf{u}$ is calculated by the following analysis problem:

$$\left.\begin{array}{lll} -\nabla \cdot (\mathbf{E} : \boldsymbol{\epsilon}(\mathbf{u})) = \mathbf{0} & \text{in} & \Omega \\ \mathbf{u} = \mathbf{0} & \text{on} & \Gamma_u \\ (\mathbf{E} : \boldsymbol{\epsilon}(\mathbf{u})) \cdot \mathbf{n} = \mathbf{t} & \text{on} & \Gamma_t \end{array}\right\}, \tag{9}$$

where $\mathbf{E}$ stands for a homogeneous isotropic elasticity tensor; $\boldsymbol{\epsilon}$ for the linear strain tensor that is defined as

$$\boldsymbol{\epsilon} = \frac{1}{2}\left(\nabla \mathbf{u} + \nabla \mathbf{u}^{\top}\right). \tag{10}$$

The material density ρ is embedded in the elasticity tensor as

$$\mathbf{E} = \rho^P \mathbf{E}_0, \tag{11}$$

where $\mathbf{E}_0$ is the elasticity tensor of the solid material, i.e., $\mathbf{E} = \mathbf{E}_0$ when $\rho = 1$; $P(= 3, \textit{typically})$ is introduced for promoting binarization within the bandwidth (not for regularizing the design space).

2.5 Algorithm for topology optimization

For solving the optimization problem (8) the main flow of the computation is summarized in Table 1. In this

Table 1 Algorithm for topology optimization

0)	Set all the constants: ϵ, h, $\underline{\phi}$, $\overline{\phi}$, $\overline{V}$, R, P.
1)	Set initial values for ϕ .
2)	Calculate φ by (3).
3)	Calculate ρ by (4).
4)	Solve the forward problem (9)
5)	Solve the adjoint problem (13).
6)	Calculate the sensitivities by (15) and (16).
7)	Update the design variables ϕ by SNOPT.
8)	Return to 2) for new iteration, or terminate.

algorithm it is assumed that a modern optimization solver SNOPT (Gill et al. 2007) is used with the first derivatives $\frac{\partial f}{\partial \phi}$ and $\frac{\partial g}{\partial \phi}$. We seek a solution that fulfill the following first order necessary condition, i.e., KKT condition:

$$\frac{\partial f}{\partial \phi} + \lambda \frac{\partial g}{\partial \phi} = 0, \quad \lambda g = 0, \quad \lambda \geq 0, \quad g \leq 0, \tag{12}$$

where λ is the Lagrange multiplier.

The sensitivity analysis of $\frac{\partial f}{\partial \phi}$ follows the standard adjoint variable method (Bendsøe and Sigmund 2003). We first solve the following adjoint problem for the adjoint variable $\tilde{\mathbf{u}}$:

$$\left.\begin{array}{lll} -\nabla \cdot \left(\mathbf{E} : \boldsymbol{\epsilon}(\tilde{\mathbf{u}})\right) = \mathbf{0} & \text{in} & \Omega \\ \tilde{\mathbf{u}} = \mathbf{0} & \text{on} & \Gamma_u \\ \left(\mathbf{E} : \boldsymbol{\epsilon}(\tilde{\mathbf{u}})\right) \cdot \mathbf{n} = -\frac{\partial f}{\partial \mathbf{u}} & \text{on} & \Gamma_t \end{array}\right\}. \tag{13}$$

In this particular case where $\frac{\partial f}{\partial \mathbf{u}} = \mathbf{t}$, therefore, $\tilde{\mathbf{u}} = -\mathbf{u}$. The sensitivity of the objective function with respect to the density is then calculated as

$$\frac{\partial f}{\partial \rho} = \boldsymbol{\epsilon}(\tilde{\mathbf{u}}) : \frac{\partial \mathbf{E}}{\partial \rho} : \boldsymbol{\epsilon}(\mathbf{u}) = -\boldsymbol{\epsilon}(\mathbf{u}) : \frac{\partial \mathbf{E}}{\partial \rho} : \boldsymbol{\epsilon}(\mathbf{u}). \tag{14}$$

With this expression and, (3) and (6) the design sensitivity can be calculated by the chain rule:

$$\frac{\partial f}{\partial \phi} = \frac{\partial f}{\partial \rho} \frac{\partial \rho}{\partial \varphi} \frac{\partial \varphi}{\partial \phi}, \tag{15}$$

where $\frac{\partial \varphi}{\partial \phi}$ corresponds to the inverse of the linear operator $(-R^2\nabla^2 + 1)^{-1}$, namely the same filter of (3) is applied to the sensitivity $\frac{\partial f}{\partial \rho}$ which is essentially non-smooth. The sensitivity $\frac{\partial g}{\partial \phi}$ can also be calculated by the chain rule as

$$\frac{\partial g}{\partial \phi} = \frac{\partial g}{\partial \rho} \frac{\partial \rho}{\partial \varphi} \frac{\partial \varphi}{\partial \phi}, \tag{16}$$

where $\frac{\partial g}{\partial \rho} = 1$.

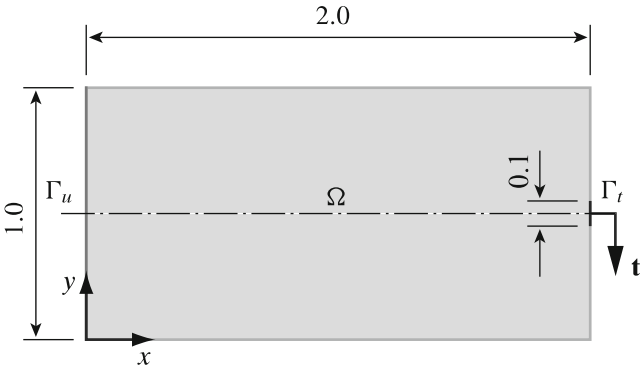


Fig. 4 Design domain and boundary conditions for the 2D minimum mean compliance problem

3 Numerical examples

The program based on the algorithm listed in Table 1 is implemented using COMSOL Multiphysics. Therefore all the formulations are discretized by the finite element method. As regards the sensitivity analysis using the software we refer to the previous work (Gersborg-Hansen et al. 2005; Olesen et al. 2006) and the user's manual.

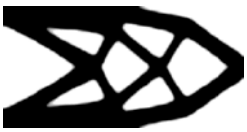
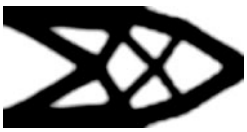
Figure 4 shows the design domain with all necessary boundary and load conditions for a benchmark example of the minimum compliance problem (8). The design domain is discretized by a square 4-node linear element with the length $\Delta x = 0.025$, thus (80×40) elements are used. Table 2 describes the parameter values adopted in the numerical examples. In this problem we pursue horizontally symmetric designs. Therefore, we halve the design domain by the horizontal centerline and apply the proposed method to the upper half of the domain. Note that we need to impose the anti-symmetric condition over the central axis when solving the analysis problem (9).

Figure 5 illustrates the optimized results by the density in grayscale when the parameter R in the Helmholtz equation (3) is set as $\{1.0\Delta x, 1.5\Delta x, 2.0\Delta x\}$, and the bandwidth of the regularized Heaviside function $2h$ as $\{0.2, 1.0, 2.0\}$, respectively. The lesser R and the tighter $2h$ (the upper left) gives finer details with a clear outline in the final design, while the opposite combination (lower right) makes the

Table 2 Parameters used for the numerical examples

Constants	Values
Lower bound for the density: ϵ	0.01
Element length: Δx	0.025
Bounds for the scalar function: $\overline{\phi}(= -\underline{\phi})$	1.0
Upper bound for the volume: $\overline{V}$	1.0

Fig. 5 Optimized results

	$R = 1.0\Delta x$	$R = 1.5\Delta x$	$R = 2.0\Delta x$
$h = 0.1$	 $f = 0.2025$, Iter.: 57	 $f = 0.2033$, Iter.: 56	 $f = 0.2042$, Iter.: 47
$h = 0.5$	 $f = 0.2095$, Iter.: 35	 $f = 0.2161$, Iter.: 45	 $f = 0.2193$, Iter.: 34
$h = 1.0$	 $f = 0.2262$, Iter.: 46	 $f = 0.2402$, Iter.: 34	 $f = 0.2501$, Iter.: 29

outline extremely blurred. Note that no checkerboard patterns have appeared in all the cases even though the linear elements are only used for the discretization.

Figure 6 shows the histories of the objective function value and the relative complementarity gap in the case where $R = 1.5\Delta x$ and $h = 0.5$. The complementarity gap indicates how well the KKT condition of (12) is fulfilled. In this particular case, assuming the j-th component of the reduced gradient as

$$d_j = \frac{\partial f}{\partial \phi_j} + \lambda \frac{\partial g}{\partial \phi_j}, \tag{17}$$

the complementarity gap e is equivalently defined as

$$e = \max_j s_j \big/ \|\lambda\|, \tag{18}$$

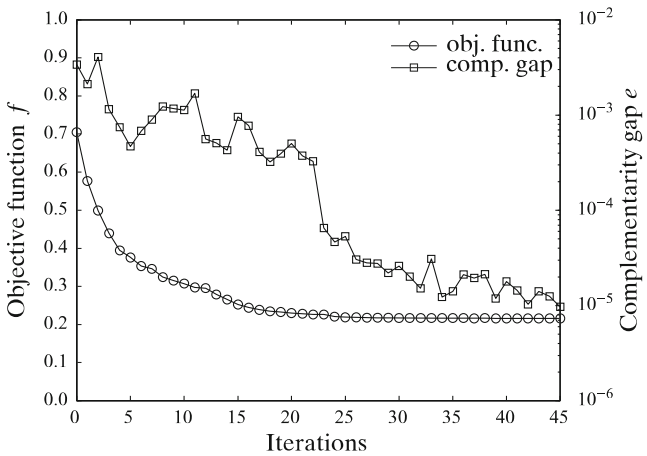


Fig. 6 Optimization histories

where

$$s_j = \begin{cases} -d_j \min \left\{ \overline{\phi} - \phi_j,\ 1 \right\} & (d_j \le 0); \\ d_j \min \left\{ \phi_j - \underline{\phi},\ 1 \right\} & (d_j > 0). \end{cases} \tag{19}$$

The optimization process is terminated when the complementarity condition is satisfied within the tolerance as $e \le 10^{-5}$ (called *optimality tolerance* in SNOPT (Gill et al. 2007)). In this case the optimization converged after 45 iterations.

4 Conclusions

An alternative method is developed for implementing a convolution filter that is applied to design variables instead of design sensitivities. Since the filter is directly applied to the design variables, the consistency of the design sensitivities is preserved. The effectiveness of the proposed method is confirmed through numerical examples.

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